

A crash-course on boosting technical skills for writing scientific reports in $\text{\LaTeX}$

Fatos Gashi

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Labels and Cross-References

- ▶ Use `\label` and `\ref` for automatic numbering.
- ▶ The `amsmath` package provides `\eqref` for referencing equations.

```
\documentclass{article}
\usepackage{amsmath} % for \eqref
\begin{document}
```

```
\section{Introduction}
\label{sec:intro}
```

In Section `\ref{sec:method}`, we `\ldots`

```
\section{Method}
\label{sec:method}
```

```
\begin{equation}
\label{eq:euler}
e^{i\pi} + 1 = 0
\end{equation}
```

By `\eqref{eq:euler}`, we have `\ldots`

```
\end{document}
```

1 Introduction

In Section 2, we ...

2 Method

$$e^{i\pi} + 1 = 0 \tag{1}$$

By (1), we have ...

Graphics

- ▶ Requires the `graphicx` package, which provides the `\includegraphics` command.
- ▶ Supported graphics formats include JPEG, PNG and PDF (usually).

```
\includegraphics[  
  width=0.5\textwidth]{gerbil}
```

```
\includegraphics[  
  width=0.3\textwidth,  
  angle=270]{gerbil}
```

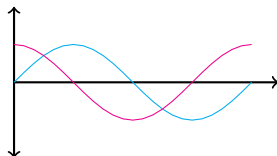


Image license: CC0

Functions

- You can even plot some simple functions.

```
\begin{tikzpicture}[scale=0.5]
% y axis
\draw[<->, thick] (0,2) -- (0,-2);
% x axis
\draw[ ->, thick] (0,0) -- (7, 0);
% curves
\draw[cyan,domain=0:2*pi]
  plot (\x, {sin(\x r)});
\draw[magenta,domain=0:2*pi]
  plot (\x, {cos(\x r)});
\end{tikzpicture}
```



Frame Title

- ▶ Rendering figures directly in $\text{\LaTeX}$
- ▶ refer to *results-plots.txt*

- Put your references in a .bib file in 'bibtex' database format:

```
@Article{Jacobson1999Towards,
  author = {Van Jacobson},
  title = {Towards the Analysis of Massive Multiplayer Online
           Role-Playing Games},
  journal = {Journal of Ubiquitous Information},
  Month = jun,
  Year = 1999,
  Volume = 6,
  Pages = {75--83}}

@InProceedings{Brooks1997Methodology,
  author = {Fredrick P. Brooks and John Kubiawicz and
           Christos Papadimitriou},
  title = {A Methodology for the Study of the
           Location-Identity Split},
  booktitle = {Proceedings of OOPSLA},
  Month = jun,
  Year = 1997}
```

- Most reference managers can export to bibtex format.

bibT_EX 2

- ▶ Each entry in the .bib file has a *key* that you can use to reference it in the document. For example, Jacobson1999Towards is the key for this article:

```
@Article{Jacobson1999Towards,  
  author = {Van Jacobson},  
  ...  
}
```

- ▶ It's a good idea to use a key based on the name, year and title.
- ▶ L^AT_EX can automatically format your in-text citations and generate a list of references; it knows most standard styles, and you can design your own.

bibT_EX 3

- ▶ Use the natbib package¹ with `\citet` and `\citep`.
- ▶ Reference `\bibliography` at the end, and specify a `\bibliographystyle`.

```
\documentclass{article}
\usepackage{natbib}
\begin{document}

\citet{Brooks1997Methodology}
show that \ldots. Clearly,
all odd numbers are prime
\citep{Jacobson1999Towards}.

\bibliography{bib-example}
% if `bib-example' is the name of
% your bib file

\bibliographystyle{plainnat}
% try changing to abbrnat

\end{document}
```

Brooks et al. [1997] show that Clearly, all odd numbers are pri
[Jacobson, 1999].

References

Fredrick P. Brooks, John Kubiawicz, and Christos Papadimitriou. A meth
ology for the study of the location-identity split. In *Proceedings of OOPSL*
June 1997.

Van Jacobson. Towards the analysis of massive multiplayer online role-playi
games. *Journal of Ubiquitous Information*, 6:75-83, June 1999.

¹There is a new package with more features named biblatex but most of
the articles templates still use natbib.

Exercise: Putting it All Together

Add an image and a bibliography to the paper from the previous exercise.

1. Download these example files to your computer.

[Click to download example image](#)

[Click to download example bib file](#)

2. Upload them to Overleaf (use the project menu).

More Neat Packages

- ▶ beamer: for presentations (like this one!)
- ▶ todonotes: comments and TODO management
- ▶ tikz: make amazing graphics
- ▶ pgfplots: create graphs in $\text{\LaTeX}$
- ▶ listings: source code printer for $\text{\LaTeX}$
- ▶ spreadtab: create spreadsheets in $\text{\LaTeX}$
- ▶ gchords, guitar: guitar chords and tabulature
- ▶ cwpuzzle: crossword puzzles

See <https://www.overleaf.com/latex/examples> and <http://texample.net> for examples of (most of) these packages.

Installing L^AT_EX

- ▶ To run L^AT_EX on your own computer, you'll want to use a L^AT_EX *distribution*. A distribution includes a latex program and (typically) several thousand packages.
 - ▶ On Windows: MikT_EX or T_EXLive
 - ▶ On Linux: T_EXLive
 - ▶ On Mac: MacT_EX
- ▶ You'll also want a text editor with L^AT_EX support. See http://en.wikipedia.org/wiki/Comparison_of_TeX_editors for a list of (many) options.
- ▶ You'll also have to know more about how latex and its related tools work — see the resources on the next slide.

Online Resources

- ▶ The Overleaf Learn Wiki — hosts these slides, more tutorials and reference material
- ▶ The $\text{\LaTeX}$ Wikibook — excellent tutorials and reference material.
- ▶ $\text{\TeX}$ Stack Exchange — ask questions and get excellent answers incredibly quickly
- ▶ $\text{\LaTeX}$ Community — a large online forum
- ▶ Comprehensive $\text{\TeX}$ Archive Network (CTAN) — over four thousand packages plus documentation
- ▶ Google will usually get you to one of the above.

Thanks, and happy T_EXing!